O-H-206B
November 9, 1976
SUPERSEDING
O-H-206a
May 12, 1965

FEDERAL SPECIFICATION

HERBICIDE, DCPA (DIMETHYL ESTER OF TETRACHLOROTEREPHTHALIC ACID)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of pre-emergence herbicide which is formulated with dimethyl ester of tetrachloroterephthalic acid in a 75 percent concentration and inert ingredients, as a water dispersible powder for application in water suspension as a spray for the selective control of crabgrass, other annual grasses, and certain broad-leaved weeds in established turf on mineral soils. The dimethyl ester of tetrachloroterephthalic acid will be referred to hereinafter in this specification as DCPA, which is the designation for this chemical adopted by the Weed Science Society of America. Further information on use is contained in section 6 of this specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

RR-S-366	-	Sieve, Test.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-D-723	-	Drums, Fiber.

Federal Standard:

FED-STD-123	-	Marking for Shipment (Civil Agencies).

FSC 6840

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the index. The index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Federal Insecticide, Fungicide, and Rodenticide Act, as amended (7 USC, Section 136 et seq.)

(Application for copies should be addressed to the Environmental Protection Agency, Pesticide Registration Division, 401 M Street, S.W., Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Composition. The herbicide powder covered by this specification shall be composed of dimethyl ester of tetrachloroterephthalic acid (DCPA), and such inert diluting modifying and conditioning agents as are needed to meet the requirements of this specification. It shall be a free-flowing powder which shall be dry, loose, and free from foreign matter. The concentration by weight of DCPA in the herbicide powder shall be not less than 75 percent when tested as specified in 4.3.3.

3.2 Chemical stability. Not more than 10 percent of the concentration of DCPA specified in 3.1 shall be lost from the herbicide when tested as specified in 4.3.3.

3.3 Reaction. The acidity of the herbicide shall be not greater than 2 percent calculated as H_2SO_4 when tested as specified in 4.3.3. The alkalinity of the powder shall be not greater than 2 percent calculated as NaOH, when tested as specified in 4.3.3.

3.4 Particle size.

3.4.1 Average diameter. The surface-mean particle diameter of the herbicide shall be not greater than 3 microns when tested as specified in 4.3.3.

3.4.2 Maximum diameter (under simulated storage conditions conducive to caking). Not less than 98 percent of the herbicide shall pass through a 74-micron (U.S. No. 200) sieve, conforming to RR-S-366, when tested as specified in 4.3.3.

3.5 Dispersibility. Not less than 95 percent of the herbicide incorporated in suspension shall pass through a 74-micron (U.S. No. 200) sieve, conforming to RR-S-366, when tested as specified for test A (without pretreatment) in 4.4.5.3.1; and not less than 90 percent shall pass through the same sieve specified herein, when tested as specified for test B (with pretreatment) in 4.4.5.3.2.

3.6 Foaming. The herbicide shall form not more than 60 mm of foam instantaneously, and not more than 25 mm of foam after standing for 5 minutes when tested as specified in 4.3.3.

3.7 Labeling. The herbicide shall be registered and labeled in compliance with the Federal Insecticide, Fungicide, and Rodenticide Act, as amended. Labels shall be approved by the Pesticide Registration Division of EPA.

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3.7.1 Label data. The label data shall include, but not be limited to, the following which shall be neatly and legibly printed on each container with black ink. The ink shall be weather-resistant and shall not smear or rub off during handling.

EPA REGISTRATION NO.
EPA ESTABLISHMENT NO.

HERBICIDE, DCPA - 75 PERCENT WETTABLE POWDER
FEDERAL SPECIFICATION O-H-206
ACTIVE INGREDIENTS
Dimethyltetrachloroterephthalate - 75%
Inert Ingredients - 25%
Total -100%
NATIONAL STOCK NUMBER
MANUFACTURER'S NAME & ADDRESS
DATE OF PACK
LOT NUMBER
CONTRACT NUMBER
NET CONTENTS

WARNING
KEEP OUT OF REACH OF CHILDREN
HARMFUL IF SWALLOWED
AVOID CONTACT WITH EYES & SKIN
IN CASE OF CONTACT, FLUSH EYES WITH PLENTY OF WATER
WASH AFTER HANDLING OR USE
AVOID CONTAMINATION OF FEED & FOODSTUFFS
STORE IN DRY PLACE
DISPOSE OF AS REQUIRED BY EPA

DIRECTION FOR USE: Apply at the rate of 14 lbs of Herbicide in 40 to 100 gallons of water per acre (1 lb/3 to 7 gals/3000 sq.ft.). Gives excellent control of crabgrass, stinkgrass, witchgrass, fall panicum, Texas millet, yellow and green foxtail lambsquarters, chickweed, redstem, filaree, purslane, and Florida pusley. This herbicide should be applied in the early spring before crabgrass seeds germinate. A second application at half the recommended rate may be made 2 months after the first application, if necessary.

3.8 Workmanship. The herbicide shall be homogeneous, contain no lumps, evidence of moisture, or presence of foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase document

4.3 End item inspection.

4.3.1 Examination of the end item. The end item shall be examined for the specific defects in 4.3.1.1 and 4.3.1.2. The applicable inspection levels and acceptable quality levels (AQLs) shall be as indicated in 4.3.2. The lot size for purposes of determining the sample size shall be in accordance with MIL-STD-105. The sample size shall be expressed in filled unit containers of the same capacity.

4.3.1.1 Examination of end item for workmanship. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Workmanship (contents)	Not homogeneous
	Contains lumps
	Any evidence of moisture
	Presence of foreign matter

4.3.1.2 Examination of filled container for net contents. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container for all sample units examined is less than specified or indicated.

4.3.2 Inspection levels and acceptable quality levels (AQLs). The inspection levels and AQLs, expressed in defects per 100 units, shall be as follows:

Examination	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	S-2	N/A

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4.3.3 Testing of the end item. The end item shall be tested for the characteristics specified in table I. The lot size shall be expressed in pounds of herbicide. The sample unit for testing shall be a 1-pound composite obtained by combining and thoroughly mixing equal portions from containers selected at random throughout the lot. The composite shall be placed in a clean, dry, amber glass, sealed container. Care shall be exercised to prevent contamination or alteration of the herbicide during the sampling, compositing, storage and testing. All test reports shall contain the individual values used in expressing the final result. The lot shall be unacceptable if the composite fails to meet any test requirements specified. The lot size and sample size shall be in accordance with the following:

Lot size (pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE I. Testing of the end item

Characteristic	Requirement paragraph	Test method	Results reported as: Pass or fail	Results reported as: Numerically to nearest
DCPA concentration	3.1	4.4.1		1 percent
Chemical stability	3.2	4.4.2		1 percent
Reaction	3.3	4.4.3		1 percent
Particle size	3.4	4.4.4	X	
Dispersibility	3.5	4.4.5	X	
Foaming	3.6	4.4.6		1 mm

4.4 Test methods.

4.4.1 Test for concentration of DCPA. Two test methods are provided herein for the determination of the concentration of DCPA, and either method may be used at the option of the manufacturer of the herbicide.

4.4.1.1 Test A: Infrared spectrophotometric method. In this method the herbicide sample is dissolved in carbon disulfide and the concentration is determined by comparing the infrared absorbance of this solution with that of a standard at a wavelength which is characteristic of DCPA.

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4.4.1.1.1 Reagents.

(a) Reagent grade carbon disulphide (CS_2).
(b) DCPA, pure spectrographic standard. (Diamond Shamrock Chemical Company, Agricultural Chemicals Division, Cleveland, OH 44115; or equivalent.

4.4.1.1.2 Apparatus.

(a) Infrared spectrophotometer, Perkin-Elmer Model 21, or equivalent.
(b) Matched sodium chloride cells of 0.5 mm path length.

4.4.1.1.3 Procedure.

(a) Weigh 0.25 g of pure DCPA standard, transfer to a 25 ml volumetric flask and bring to volume with CS_2. Allow sufficient time for the herbicide standard to dissolve completely (30 minutes to 1 hour).
(b) Weigh 0.25 g of sample and bring to volume in 25 ml volumetric flask with CS_2. Since some CS_2 insoluble material may be encountered, allow sample to stand same length of time as standard.
(c) With CS_2 in reference cell and the standard solution in sample cell, scan from 9.5 to 11.5 microns under the following instrumental conditions:

Source	0.3 amps
Slit	180 microns (manual)
Gain	optimum
Response	1 : 1
Pen	set at 0 percent transmission with slight upscale drift applied
Pen speed	3 seconds full scale
Scanning speed	2.5 min./micron
Region scanned	9.5 to 11.5 microns
Analytical wavelength	10.4 microns

(d) Whatever the instrument and operating parameters used, it is important that both standard and sample are scanned under identical conditions.
(e) Upon completion of standard and sample scans, a base line is drawn from 9.3 to 11.4 microns and the absorbency obtained at 10.4 microns.
(f) Calculate the percent DCPA from following relationship:

As (DCPA std.): grams DCPA std:: As (sample): x grams DCPA in sample

$$\frac{\text{grams of DCPA in sample} \times 100}{\text{Total Sample weight}} = \text{percent DCPA}$$

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(g) Under the instrumental conditions outlined above, the concentrations of standard and sample used will cause the band at 10.4 microns to approach optimum percent transmission.

4.4.1.2 Test B: Gas chromatographic method. This method consists of the extraction of the DCPA from the formulation with acetone, the removal of interfering materials by column chromatography, and the analysis of DCPA by means of vapor phase chromatography (VPC). The DCPA is extracted from the inert component using a soxhlet apparatus. The extracted DCPA is then passed through a florisil column to remove all of the interfering materials. The DCPA is concentrated in carbon tetrachloride to a volume of 25 ml. This sample and a standard of about the same concentration are chromatographed. From this analysis the percent DCPA in the original formulation can be calculated.

4.4.1.2.1 Reagents.

(a) Carbon tetrachloride, reagent grade.
(b) Acetone, reagent grade.
(c) Florisil, 60/100 mesh. (Floriden Company, Tallahassee, FL).
(d) Pure DCPA standard.

4.4.1.2.2 Apparatus.

(a) Soxhlet extraction apparatus (size B, thimble 33 x 94 mm).
(b) Chromatographic unit with proper column, temperature, and flow. (Burrell Kromatog K-1 or K-2).
(c) Analytical balance.
(d) 250 ml flat bottom flasks 24/40 ground glass joints.
(e) 250 ml Griffin beakers, with spout.
(f) 250 ml graduated cylinders.
(g) Glass wool. (Corning Glass Works Cat. No. 3950).
(h) Steam bath.
(i) Boiling chips - Berl saddles (Fisher 9-191-5, size 6 mm).
(j) 25 ml volumetric flasks with glass stoppers.
(k) Ring stands and clamps.
(l) Clean up columns (absorption chromatographic columns - 22 x 500 mm).
(m) Hot plate.

4.4.1.2.3 Procedure.

Extraction:

(a) Accurately weigh and transfer to an extraction thimble, sufficient herbicide sample to obtain 0.5 to 1.0 g of pure DCPA. This will require 30 to 50 g of a 1.5 percent formulation. (For formulations containing a higher percentage of DCPA it is desirable to extract at least 10 to 20 g of the formulation, dilute to a known volume with acetone, and take an aliquot equivalent to 0.5 to 1.0 g of DCPA).

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(b) Cover the surface of the sample with a plug of glass wool and place the thimble in the extraction apparatus.

(c) Add 150 ml of acetone to the 250 ml extraction flask. Also, add two boiling chips to ensure smooth boiling.

(d) Assemble the extraction apparatus and place on a hot plate. Supply sufficient heat to maintain a steady discharge of solvent from the condenser.

(e) Continue to extract for 8 hours.

(f) When the extraction has been completed, remove the flask. If the total extract contains less than 1 g of DCPA, evaporate the solvent to a volume of about 50 ml on the steam bath. If the laboratory is equipped with compressed air outlets, a stream of clean, dry air can be used to aid in the solvent removal. If the extract contains more than 1 g of DCPA, dilute to a known volume, take a suitable size aliquot for analysis and concentrate the solvent to a volume of about 50 ml.

(g) Add 25 ml of carbon tetrachloride to the flask and concentrate the solvent to a volume of about 20 ml.

Cleanup:

(a) Prepare the glass columns for the cleanup of the extracted materials by first pushing a piece of glass wool to the bottom of the column with a glass rod, then filling the column to a height of four inches with florisil (packing with the glass rod every inch).

(b) Condition the column by filling it with carbon tetrachloride, allowing the carbon tetrachloride to flow through the florisil to remove any soluble material and to prepare the column for a fast, even flow.

(c) Quantitatively transfer the contents of the 250 ml flask into the column and allow the meniscus of the solution to reach the top of the florisil before adding wash solution.

(d) Wash the column with 10 ml of a solution of 5 percent acetone in carbon tetrachloride and allow the meniscus of the mixture to reach the top of the florisil. Repeat three times.

(e) Add 120 ml of the 5 percent acetone solution to the column to completely elute all the DCPA from the column.

(f) The eluent from the column is collected in a 250 ml beaker and concentrated to a volume of 10 ml on a steam bath using a current of clean, dry air.

(g) Quantitatively transfer the 10 ml. of eluent from the beaker to a 25 ml volumetric flask and dilute to the mark with carbon tetrachloride.

Analysis:

(a) Conditions for the Burrell Kromatog are:

(1) Column - 2.5 meter glass hair pin column packed with 18 percent Dow Corning silicone stopcock grease on fire brick (Kromat FB). Temperature 213° C. (voltage of 80).

(2) Detector temperature 250°C. Current 250 ma.

(3) He flow 54 ml per minute. Ten units back pressure.

(4) Attenuator set at 2.

(Any chromatographic instrument with a suitable column and conditions would be satisfactory).

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Analysis: (cont'd)

(b) Prepare a standard by weighing accurately 1.0000 g ± 0.0005 g of pure DCPA into a 25 ml volumetric flask and diluting to the mark with carbon tetrachloride.

(c) Using a 10 microliter syringe, withdraw exactly 5.0 microliters of the standard solution from the volumetric flask and inject it into the column.

(d) Integrate the peak corresponding to the pure DCPA. Repeat the injections until the areas check within 1 percent.

(e) Clean the syringe and withdraw 5.0 microliters of solution from the herbicide sample and inject it into the column in the same manner as the standard.

(f) Integrate the area of the DCPA herbicide in the sample. Repeat injections until the area checks are within 1 percent.

Calculation:

$$\text{Grams DCPA in std.} \times \frac{\text{area sample (avg.)}}{\text{area std. (avg.)}}$$

$$\times \frac{100}{\text{wt. of sample extracted}} = \text{Percent DCPA}$$

4.4.2 Chemical stability. This test is intended for the determination of any chemical breakdown of the herbicide when stored at a high temperature.

4.4.2.1 Apparatus.

(a) Oven, regulated to maintain a temperature of 50° ± 2°C.

(b) Storage jars: glass, 60 to 120 ml, screw-cap with aluminum inserts in caps.

4.4.2.2 Procedure. Load a clean, dry storage jar approximately half full of the product without compacting. Close the jar tightly with the aluminum insert cap, and store in the oven at 50° ± 2°C for 14 days. Remove the jar from the oven and allow to cool to room temperature. Weigh out a one gram sample, and analyze for DCPA as described in 4.4.1. Calculate the loss of DCPA, if any, in terms of percentage of DCPA originally present.

4.4.3 Reaction.

4.4.3.1 Procedure to determine acidity. Weigh exactly 10 g of the sample and disperse in 100 ml of distilled water, stirring frequently for 1/2 hour. Filter and titrate the filtrate immediately with 0.02 N sodium hydroxide (NaOH), using methyl red as indicator. Carry out a blank determination on 100 ml distilled water with 0.02 N NaOH.

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Calculation:

Acidity as H_2SO_4 (percent) = 0.0098 X (a - b)
where a = volume in ml of 0.02 NaOH used for the sample
b = volume in ml of 0.02 NaOH for the blank

NOTE: The blank may take the form of a small titre with 0.02 N HCl, in which case

Acidity at H_2SO_4 (percent) = 0.0098 X (a + c)
where a = volume in ml of 0.02 N NaOH used for the sample
c = volume in ml of 0.02 N HCl used for the blank
Alternatively, the end-point may be determined electrometrically.

4.4.3.2 Procedure to determine alkalinity. Weigh exactly 10 g of the herbicide sample and disperse in 100 ml of distilled water, stirring frequently for 1/2 hour. Filter, and titrate the filtrate immediately with 0.02 N hydrochloric acid (HCl), using methyl red as indicator. Carry out a blank determination on 100 ml of distilled water with 0.02 N NaOH.

Calculation:

Alkalinity as NaOH (percent) = 0.008 X (d + e)
where d = volume in ml of 0.02 N HCl used for the sample
e = volume in ml of 0.02 N NaOH used for the blank

NOTE: The blank may take the form of a small titre with 0.02 N HCl, in which case

Alkalinity as NaOH (percent) = 0.008 X (d - f)
where d = volume in ml of 0.02 N HCl used for the sample
f = volume in ml of 0.02 N HCl used for the blank
Alternatively, the end-point may be determined electrometrically.

4.4.4 Particle size.

4.4.4.1 Determination of average diameter. The surface-mean particle diameter shall be determined by the air-permeation method. The apparatus used shall be a Government-approved air-permeation apparatus (see 6.3). The solid density of the herbicide power, knowledge of which is prerequisite to particle fineness measurement by the air-permeation procedure, may be determined by any standard method. A suitable method is the conventional method of liquid immersion in a specific gravity bottle having a ground-in thermometer, with distilled water as the immersion liquid. In the use of this method, care must be taken to guard against significant loss of water-soluble material; the pycnometer (specific gravity bottle) should have a nominal volume of at least 25 ml, and the weight of sample used for observation should be as great as can be handled conveniently in the pycnometer used.

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4.4.4.2 Determination of maximum diameter.

4.4.4.2.1 Apparatus.

(a) Pretreatment jar; an air-tight screw-cap glass jar, 35 ml ± 5 ml actual capacity.

(b) Oven; regulated to maintain 55° ± 1°C.

(c) Mixing jar; a screw-cap glass jar, 295 to 300 ml (10 U.S. fl. oz.), substantially cylindrical, approximately 8 cm deep, without abrupt shoulders. (Some kinds of conventional fruit jars quality.)

(d) Tumbling machine; any mechanical device for rotating the mixing jar around its center, end-over-end, at 60 r.p.m.

(e) Sieve; a 74-micron (U.S. No. 200) standard testing sieve, 8 inch frame diameter, conforming to RR-S-366.

(f) Sprayer; consisting of a shower head, with not less than 50 perforations distributed practically uniformly over a circular faceplate area 6 to 8 cm in diameter, connected by a hose to a source of tap water. The relation of water pressure to size and number of perforations must be such that if the shower head be inverted to send a spray upward, no considerable portion of the spray will rise more than 15 cm above the faceplate. If the pressure as limited by this criterion is insufficient to give the prescribed flow rate (see subparagraph (3), the flow rate must be adjusted by increasing the size and/or number of perforations.

4.4.4.2.2 Procedure. This test, which is primarily a sieve test designed so that essentially all the material in an acceptable sample will pass through the sieve, involves a pretreatment to simulate storage conditions conducive to caking. The complete test procedure consists of three steps: (1) the pretreatment; (2) the preparation of a suspension of the treated sample in water; and (3) the actual screening and determining the percentage of residue. The three steps shall be performed in the following manner:

(1) Pretreatment. Store the opened jar and screw-cap at 55°C for at least a 1-hour period; then, without allowing the jar to cool, pour a 2 g sample of the powder into the jar, seal the jar airtight and keep it in the oven at 55° ± 1°C for 24 hours. Remove from oven and let stand at room temperature, still sealed, for 3 hours; then proceed promptly to preparation of suspension as follows:

(2) Preparation of suspension. (a) Deposit the 2 g sample of pretreated powder on a 50 ml portion of distilled water in the mixing jar and let stand for 3 minutes undisturbed. (b) Close the jar and tumble it on the machine at 60 r.p.m. for 5 minutes. (c) Dilute with 50 ml of distilled water. (d) Tumble as before, for 1 minute, and proceed as follows:

(3) Screening. Transfer the suspension to the sieve which, in preparation for the test must have been freed of any film of grease or other water-repellent material. Wash the residue on the sieve for 10 minutes with the sprayer adjusted to deliver 4 to 5 liters per minute. Transfer the residue to a filter whose weight has been determined on an oven-dried basis (oven temperature not over 80°C) dry the residue in an oven at the same temperature as was used for the tare weight; and calculate the weight percentage of sample passing through the sieve.

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4.4.5 Dispersibility.

4.4.5.1 Reagents. Hard water; designed to have a hardness of 342 p.p.m. calculated as calcium carbonate; composition as follows:

Calcium chloride anhydrous	0.304 gram
Magnesium chloride hexahydrate	0.139 gram
Distilled water to make	1 liter

4.4.5.2 Apparatus. Same as for maximum diameter test (see 4.4.4.2.1) with the omission of sprayer.

4.4.5.3 Procedure. The dispersibility of the herbicide powder in water is tested by the following special sieve tests in which the suspension is simply poured upon the sieve without any washing or any agitation of the sieve. In preparation for these tests, make sure that the wire cloth of the sieve is free from any film of grease or other water-repellent material and is thoroughly wetted with water on both sides and kept so wetted until the moment for use.

4.4.5.3.1 Test A: Without pretreatment. Prepare a suspension as specified in subparagraph (2) of 4.4.4.2.2 except to omit the pretreatment of the herbicide powder and to substitute hard water for distilled water throughout. When the final tumbling is completed, remove the jar from the machine, and with the sieve standing in a level position, pour the suspension as evenly as possible over its whole area, taking care meanwhile not to allow any powder to settle in the jar. Raise one side of the sieve frame 25 mm by means of a prop, and allow the residue to drain undisturbed for 5 minutes. Transfer the residue to a filter, dry and weigh as in subparagraph (3) of 4.4.4.2.2, and calculate the dispersibility at 74-microns as weight percentage of sample passing through the sieve. (Filtration may often be facilitated by washing the residue from the sieve into a beaker, adding a few ml. of dilute alum solution to flocculate the solids, then pouring onto a coarse filter).

4.4.5.3.2 Test B: With pretreatment. The procedure is the same as for test A, except that the pretreatment specified in subparagraph (1) of 4.4.4.2.2 is included.

4.4.6 Foaming.

4.4.6.1 Apparatus. The apparatus is shown in figure 1 and is made to standard dimensions from stock glass. The apparatus consists of a reservoir or pipette (A), a receiver (B), a jacket (C), and a supporting stand and other parts required to make a complete assembly. The orifice of the reservoir is so constructed that the turbulence set up causes the stream of solution to break up into droplets just after emerging from the orifice. Foam is produced when these droplets strike the surface of the liquid or foam in the receiver.

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Construct the tip of the reservoir from pyrex glass tubing whose outside diameter is approximately 7 mm, and length is 6 cm. Select another piece of tubing whose outside diameter is such that it fits snugly inside the 7 mm piece. From this cut a short 1 cm section and grind it square at both ends. Place this inner section flush with the extreme tip of the large piece, and then heat the outside of the larger tube to redness at one or two points opposite the middle of the inner section. Join the tip assembly to the reservoir proper.

Orifices varying in size from 2 mm to 3 mm in diameter produce nearly the same amount of foam. In standardizing several pieces of apparatus to conform to each other, it is permissible to alter slightly the solution volume in the reservoir.

4.4.6.2 Procedure. Prepare a suspension in the ratio of 45.36 g active ingredient in 1 gallon of hard water (see 4.4.5.1) at 18° to 24°C.

To set up and operate the test, proceed as follows:

(1) Adjust the receiver so that it is plumb.

(2) Adjust the reservoir so that a stream of water emerging from the orifice will strike the bottom center of the receiver.

(3) Be sure that both the receiver and reservoir are clean. The final step in the cleansing shall be a thorough rinsing with distilled water. If the glass surfaces are clean, the distilled water should flow over the side walls in an unbroken film, finally draining without the appearance of droplets on the glass.

(4) Circulation shall be maintained through the jacket at a controlled temperature of 18° to 24°C.

(5) Close the stopcock of the receiver, and by means of a 50 ml volumetric pipette introduce 50 ml of the test suspension into the receiver. The suspension should be run down the side walls of the receiver to avoid foam formation and to wet the entire interior of the receiver. Hold the tip of the pipette against the side wall of the receiver, and slowly move the tip around the circumference of the tube so that the suspension will flow down on all sides.

(6) Fill the reservoir in the manner of filling a volumetric pipette. The reservoir should be marked at a point which indicates a liquid content of 200 ml.

(7) Place the reservoir in position at the head of the receiver, and open the reservoir stopcock.

(8) When all the liquid has run out of the reservoir, start stopwatch and take foam height readings immediately and after 5 minutes. The first reading is the accepted standard, but additional indications of the relative stability of various foams may be obtained in those cases where breakdown does occur in five minutes.

(9) The readings should be taken as follows: A millimeter scale is placed behind the receiver so that the zero mark is opposite the point in the receiver which would be reached by the total volume of liquid introduced into the receiver. The foam production is measured at the top of the foam column at the highest average height to which the rim of the foam has reached. This height is proportional to the volume of air remaining in the foam. Additional measurements of foam height may be taken at suitable intervals.

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4.5 Examination of preparation for delivery. The packaging, packing, and marking of the herbicide shall be inspected to determine conformance to the section 5 requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The herbicide shall be packaged in 1 pound or 5 pound quantities in metal cans conforming to type V, class 2, exterior coating plan B of PPP-C-96, or in 50 pound quantities in fiber drums conforming to type III, grade A of PPP-D-723, as specified (see 6.2).

5.1.2 Level C. The herbicide shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The size of containers shall be as specified (see 6.2). The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The herbicide packaged in 1 pound or 5 pound metal cans as specified in 5.1.1 shall be packed in accordance with the level A packing requirements as specified in the appendix of PPP-C-96. The herbicide packaged in 50 pound fiber drums as specified in 5.1.1 requires no overpacking.

5.2.2 Level B. The herbicide packaged in 1 pound or 5 pound metal cans as specified in 5.1.1 shall be packed in accordance with the level B packing requirements as specified in the appendix of PPP-C-96. The herbicide packaged in 50 pound fiber drums as specified in 5.1.1 requires no overpacking.

5.2.3 Level C. The herbicide shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers and packing shall comply with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. Marking shall comply with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to the requirements of 3.7 and to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military agencies. In addition to the requirements of 3.7 and to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2.1 Military marking. The following weeds are grouped according to their susceptibility to Herbicide DCPA:

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Moderately Susceptible (higher rates required for effective control): Barnyard grass, goose grass, annual blue-grass, Johnson-grass, browntop panicum, Palmer amaranth, redroot pigweed, spiny pigweed, spotted spurge, prostrate spurge, Wright ground-cherry and Sesbania.

Tolerant: Cheat, downy bromegrass, wild oats, velvet-leaf, ragweed, mustards, rutabaga, Jimsonweed, Galinsoga, smartweed, dandelions and plantain.

6. NOTES

6.1 Intended use. The pre-emergence herbicide covered by this specification is intended for use in the selective control of crabgrass, other annual grasses, and certain broad-leaved weeds in established turf. It may thin fine-leaved fescues and bent grasses, and it reduces the germination of turf-grass seeds if sown within 60 days after application of the herbicide.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Size of container (see 5.1.1 and 5.1.2).
(d) Marking required (see 5.3).

6.3 Air permeation apparatus. An approved apparatus for use in conducting the particle size test in 4.4.4.1 is the self-calculating air-permeation apparatus of the U.S. Department of Agriculture (Patent No. 2,261,802), or equal. The Sub-Sieve Sizer, manufactured by the Fisher Scientific Company, may be used as the test instrument. Each individual apparatus is subject to inspection by the agency procuring herbicide in order to verify calibration and performance.

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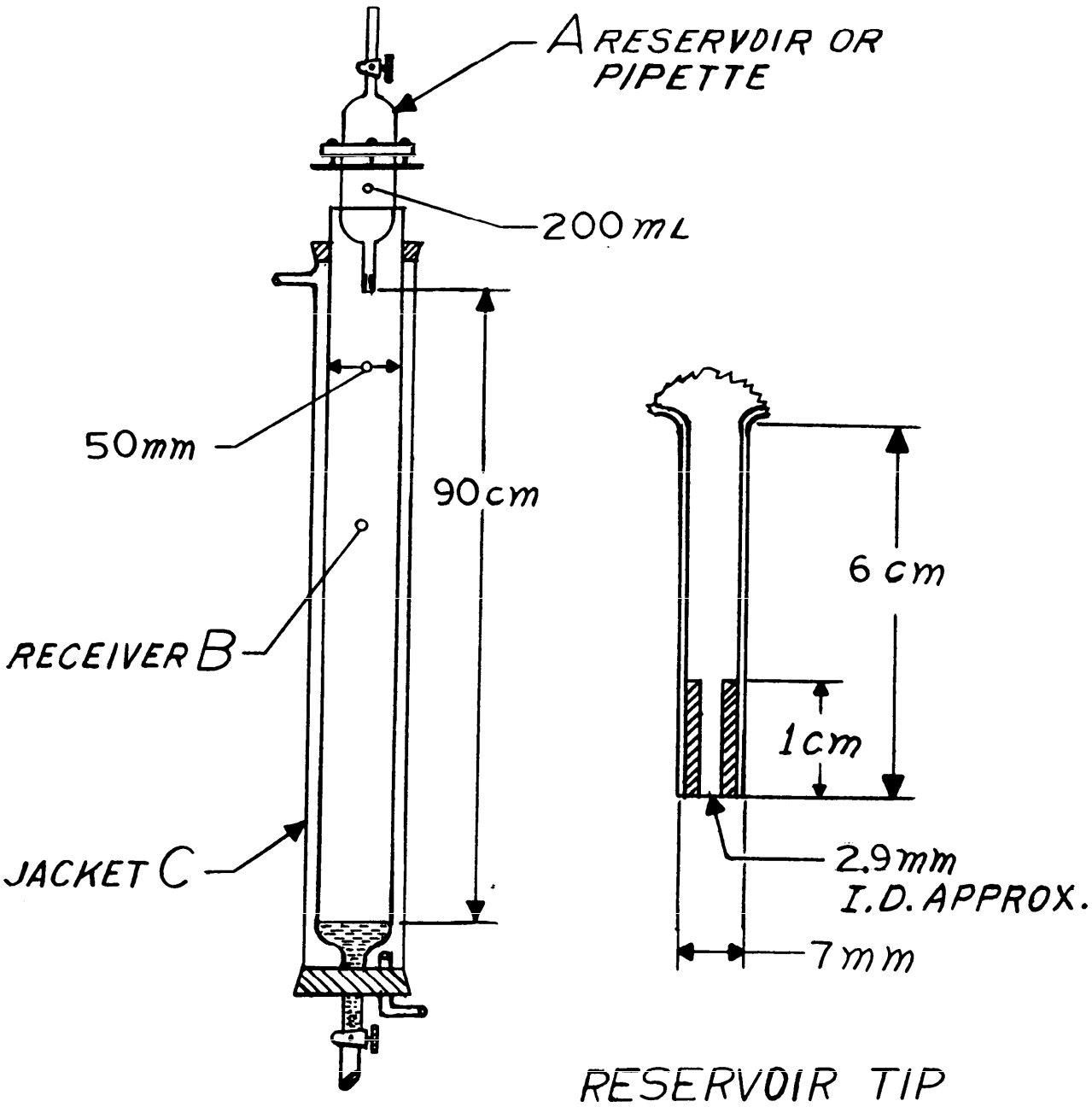


FIG 1- FOAMING TEST APPARATUS

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MILITARY CUSTODIANS:

Army - GL
Navy - YD
Air Force - 68

Review activities:

Army - MD
Navy - MS

User activities:

Army - SM
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

AGR - AFS
DOT - SMS
GSA - FSS
HEW - FDA, NIH

PREPARING ACTIVITY:

Army - GL

Project No. 6840-0306

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